SET

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Equilibrium uniqueness and representation

In nonlinear models, standard sufficient conditions for uniqueness can be difficult to verify.

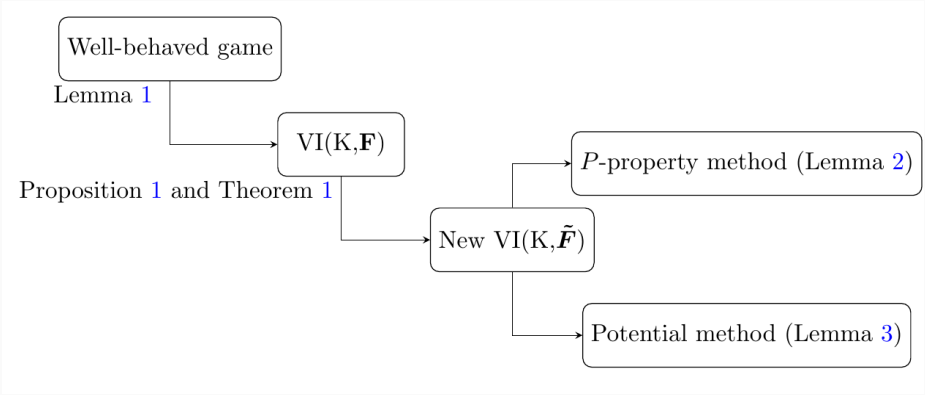
Some of this difficulty may come from how the equilibrium conditions are written.

The central question

How can we transform nonlinear equilibrium to simplify analysis, without changing the equilibrium set?

The paper develops sign-equivalent transformations (SET) to reveal useful mathematical structure in equilibrium problems.

The paper's map



An equilibrium is a best-response fixed point. The figure follows its VI representation through a sign-equivalent transformation to two methods for studying uniqueness.

Example 1: a two-player game

Two players choose $x_i \geq 0$, so $K = \mathbb{R}^2_+$.

Payoffs

$$u_1(x) = x_1\left(1 + \tfrac{1}{2}x_2\right) - \tfrac{1}{2}x_1^2,$$
$$u_2(x) = 4\left[x_2\left(1 + \tfrac{1}{2}x_1\right) - \tfrac{1}{2}x_2^2\right].$$

Each payoff is strictly concave in the player's own action.

Best responses

$$B_1(x_2) = 1 + \tfrac{1}{2}x_2, \qquad B_2(x_1) = 1 + \tfrac{1}{2}x_1.$$

The factor 4 changes player 2's payoff units, but leaves the best response unchanged.

Example 1: the VI and its transformation

For a candidate $x^* \in K$, use the negative marginal payoffs:

$$F_i = -\partial_i u_i, \qquad F(x) = \begin{pmatrix} x_1 - 1 - \frac{1}{2}x_2 \\ 4(x_2 - 1 - \frac{1}{2}x_1) \end{pmatrix}.$$

$$x^* \in \mathrm{NE} \iff \langle y - x^*, F(x^*) \rangle \geq 0 \quad \text{for every } y \in K.$$

SET: divide the second component by 4

$$\widetilde{F}(x) = \mathrm{diag}(1, \tfrac{1}{4})F(x) = \begin{pmatrix} x_1 - 1 - \frac{1}{2}x_2 \\ x_2 - 1 - \frac{1}{2}x_1 \end{pmatrix}.$$

Positive scaling preserves every component's sign on K. Since K is rectangular,

$$\mathrm{Sol}(K, F) = \mathrm{Sol}(K, \widetilde{F}).$$

Example 1: the two uniqueness routes

The transformed system has a symmetric derivative:

$$H = D\widetilde{F} = \begin{pmatrix} 1 & -\frac{1}{2} \\ -\frac{1}{2} & 1 \end{pmatrix}, \qquad \lambda(H) \in \{\tfrac{1}{2}, \tfrac{3}{2}\}.$$

P-property method

The principal minors of H are $1, 1, \frac{3}{4} > 0$.

The P-matrix condition gives at most one VI solution.

Potential method

$$\Phi(x) = \tfrac{1}{2}(x_1^2 + x_2^2) - \tfrac{1}{2}x_1x_2 - x_1 - x_2.$$

$\nabla\Phi = \widetilde{F}$ and $\nabla^2\Phi = H \succ 0$. The VI minimizes the strictly convex Φ over K.

$$\widetilde{F}(x^*) = 0 \quad \Longrightarrow \quad \boxed{x_1^* = x_2^* = 2}.$$

This feasible zero proves existence. The equilibrium is unique.

Example 2: different nonlinear best responses

Two players choose $x_i \geq 0$, with $K = \mathbb{R}^2_+$.

Payoffs

$$u_1(x) = x_1(1 + x_2) - x_1^2 - \tfrac{1}{3}x_1^3,$$
$$u_2(x) = x_2 \ln(2 + x_1) - \tfrac{1}{2}x_2^2.$$

Each payoff is strictly concave in the player's own action.

Best responses

$$B_1(x_2) = \sqrt{2 + x_2} - 1, \qquad B_2(x_1) = \ln(2 + x_1).$$

Every equilibrium is interior because both best responses are positive on K.

Example 2: a nonlinear sign-equivalent transformation

The negative marginal payoffs define $F_i = -\partial_i u_i$.

Original VI

$$F(x) = \begin{pmatrix} x_1^2 + 2x_1 - 1 - x_2 \\ x_2 - \ln(2 + x_1) \end{pmatrix}.$$

$$\partial_2 F_1 = -1, \quad \partial_1 F_2 = -\frac{1}{2 + x_1}.$$

After SET

$$\widetilde{F}(x) = \begin{pmatrix} x_1^2 + 2x_1 - 1 - x_2 \\ e^{x_2} - 2 - x_1 \end{pmatrix}.$$

$$\partial_2 \widetilde{F}_1 = \partial_1 \widetilde{F}_2 = -1.$$

The exponential is strictly increasing:

$$x_2 \gtreqless \ln(2 + x_1) \quad \Longleftrightarrow \quad e^{x_2} \gtreqless 2 + x_1.$$

All component signs agree on the rectangular domain K. Thus

$$\mathrm{NE} = \mathrm{Sol}(K, F) = \mathrm{Sol}(K, \widetilde{F}).$$

Example 2: a convex equilibrium problem

The transformed operator is a gradient, $\widetilde{F} = \nabla\Phi$, where

$$\Phi(x) = \tfrac{1}{3}x_1^3 + x_1^2 - x_1 - x_1x_2 + e^{x_2} - 2x_2.$$

For every $x \in K$,

$$\nabla^2\Phi(x) = \begin{pmatrix} 2 + 2x_1 & -1 \\ -1 & e^{x_2} \end{pmatrix} \succeq \begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix} \succ 0.$$

Unique equilibrium

Strong convexity and coercivity give a unique minimizer on K.

SET exposes a convex optimization representation of the same equilibrium problem.

Applications of general interest

1. **Market systems:** production networks, nonlinear peers, and oligopoly pricing.
2. **Nonlinear network complements:** when equilibrium exists, who acts most, and who is the key player?
3. **Network public goods:** how convex costs change participation, uniqueness, and effort rankings.

Only one methodological fact is needed before the examples.

The one methodological fact we need

Sign-equivalent transformation

If every component of F and $\widetilde{F}$ has the same sign at every feasible profile, then on a product of individual intervals the two variational inequalities have exactly the same solution set.

Player's choice	Equilibrium condition	What matters
Lower bound	$F_i \geq 0$	Sign
Interior	$F_i = 0$	Zero
Upper bound	$F_i \leq 0$	Sign

The product-domain condition matters when a choice is at a boundary.

What a better representation can reveal

P-property route

A transformed derivative matrix can establish at most one equilibrium. **Used for:** production prices, peer effects, oligopoly pricing, CES complements.

Potential route

A symmetric transformed system can become an optimization problem; curvature can give uniqueness. **Used for:** network games, public goods, multiplex interactions.

Existence needs its own argument in each application.

Production networks: the price question

An industry's output is another industry's input.

- Industry i hires labor at wage w and buys goods from its suppliers.
- Its productivity is $z_i > 0$. Its output sells at price $p_i > 0$.
- Firms take prices as given and have constant returns to scale.

A cheaper supplier lowers its customers' costs. Their lower prices can then feed back into the original supplier's costs.

The equilibrium question

For a given wage, which prices are consistent with every active industry's cost of production? Can there be more than one such price vector?

We study this price subsystem, holding the wage fixed.

A firm's cost of production

The technology is Cobb–Douglas between labor and an intermediate-input bundle. Let $Q_i(p)$ be the price of one unit of that bundle.

Cost of producing one unit

After choosing the cheapest input quantities, the firm has unit cost

$$c_i(p, w) = \frac{w^{\alpha_i} Q_i(p)^{1-\alpha_i}}{z_i}.$$

- $\alpha_i \in (0, 1)$ is labor's cost share. Intermediates account for $1 - \alpha_i$.
- Higher wages or supplier prices raise unit cost.
- Doubling z_i halves unit cost, holding input prices fixed.

Constant returns make unit cost independent of the amount produced.

Two industries: a simple price system

Measure prices in units of the fixed wage, so $w = 1$. Each industry spends half its cost on labor and half on the other industry's good.

$$Q_1 = p_2, \qquad Q_2 = p_1, \qquad \alpha_1 = \alpha_2 = \tfrac{1}{2}.$$

Competition and constant returns require price = unit cost for an industry producing a positive amount:

$$\boxed{p_1 = \frac{\sqrt{p_2}}{z_1}, \qquad p_2 = \frac{\sqrt{p_1}}{z_2}}.$$

- A price below unit cost cannot sustain positive production.
- A price above unit cost would generate unbounded profit at given prices.

At $z_1 = z_2 = 1$, the prices $p_1 = p_2 = 1$ satisfy both equations.

Two industries: a productivity shock

Industry 1's productivity doubles: $z_1 = 2$, $z_2 = 1$. Take exact logs, with $x_i = \log p_i$:

$$x_1 = -\log 2 + \tfrac{1}{2}x_2, \qquad x_2 = \tfrac{1}{2}x_1.$$

Since $x_1 = -\log 2 + \frac{1}{4}x_1$,

$$x_1 = -\tfrac{4}{3}\log 2, \qquad x_2 = -\tfrac{2}{3}\log 2.$$

	Before the shock	After the shock
Industry 1's price	1	$2^{-4/3} \approx 0.397$
Industry 2's price	1	$2^{-2/3} \approx 0.630$

If p_2 stayed at 1, then p_1 would be 0.5. Input feedback lowers it further to 0.397.

Many suppliers: the CES input price index

With several suppliers, the cheapest intermediate bundle has unit price

$$Q_i(p) = \left(\sum_j \omega_{ij} p_j^{1-\sigma_i} \right)^{1/(1-\sigma_i)}, \qquad \omega_{ij} \geq 0, \quad \sum_j \omega_{ij} = 1.$$

- ω_{ij} is supplier j's technology weight.
- $\sigma_i > 1$ is the elasticity of substitution among suppliers.

The same zero-profit condition

$$p_i = \frac{Q_i(p)^{1-\alpha_i}}{z_i} \qquad (w = 1).$$

With several suppliers, these coupled equations are generally nonlinear.

SET: the same prices in a better representation

Write $x_i = \log p_i$, $\varepsilon_i = \log z_i$, and $p = e^x$.

$$p_i z_i = Q_i(p)^{1-\alpha_i} \quad \Longleftrightarrow \quad x_i = \underbrace{-\varepsilon_i + (1-\alpha_i)\log Q_i(e^x)}_{T_i(x)}.$$

Why this transformation preserves equilibrium

The logarithm is strictly increasing. For every positive price vector,

$$\operatorname{sgn}\left[p_i z_i - Q_i(p)^{1-\alpha_i}\right] = \operatorname{sgn}[x_i - T_i(x)].$$

The change of variables $p = e^x$ is one-to-one between $\mathbb{R}^n$ and positive prices. Thus $x = T(x)$ describes exactly the same positive price equilibria.

The next step is to examine how much one industry's log price affects another's.

Labor limits the feedback in prices

Shares in total cost measure log-price pass-through:

$$W_{ij}(x) = \frac{\partial T_i(x)}{\partial x_j} \geq 0, \qquad \sum_j W_{ij}(x) = 1 - \alpha_i < 1.$$

Labor's price stays fixed. A common 1% rise in suppliers' prices raises unit cost by less than 1% to first order.

A unique positive price vector

Let $q = \max_i(1 - \alpha_i) < 1$. Then

$$\|T(x) - T(y)\|_\infty \leq q\|x - y\|_\infty.$$

The contraction theorem gives existence and uniqueness on $\mathbb{R}^n$.

The paper's P-matrix proof uses the same cost shares.

Cobb–Douglas: tracing the network multiplier

As $\sigma_i \to 1$, supplier cost shares become constant:

$$a_{ij} = (1 - \alpha_i)\omega_{ij}, \qquad A = (a_{ij}), \qquad x = -\varepsilon + Ax.$$

Direct and indirect productivity effects

$$x^* = -(I - A)^{-1}\varepsilon, \qquad (I - A)^{-1} = I + A + A^2 + \cdots.$$

- I: a productivity gain lowers the producer's own price.
- $A, A^2, \ldots$: cheaper inputs transmit the gain through successive supply links.

In our example, each link has weight $\frac{1}{2}$, so a full feedback loop has weight $\frac{1}{4}$. The same inverse appears in Bonacich centrality, with $A = \delta G$ and a different forcing vector.

Existing application: nonlinear peer effects

A social norm need not be the average peer

$$\widetilde{x}_{-i}(\beta) = \left(\sum_j \widetilde{g}_{ij} x_j^{\beta} \right)^{1/\beta}, \qquad x_i = (1-\lambda_2) a_i + (\lambda_1 + \lambda_2) \widetilde{x}_{-i}(\beta).$$

- $\beta = 1$ recovers linear-in-means; other values change whose behavior is most influential.
- Taking logs exposes a *P*-property: under the paper's parameter restrictions, any solution is unique.
- The result is conditional on existence; the transformation alone does not supply it.

Existing application: oligopoly with price bounds

Firm i chooses $p_i \in [a_i, b_i]$ and earns $(p_i - c_i)Q_i(p)$.

A transformed pricing condition

$$F_i = -\frac{\partial \pi_i}{\partial p_i} \quad \sim_{\text{sign}} \quad \widetilde{F}_i = p_i - c_i + \frac{Q_i}{\partial Q_i / \partial p_i}.$$

The positive rescaling preserves the signs at both interior prices and binding price floors or caps.

- Under the paper's demand conditions E1–E3, the transformed system has the P-property.
- Therefore the Bertrand game has at most one pure-strategy equilibrium.

Existing applications: network games and multiplexity

Network public goods with income

An inverse Engel curve turns heterogeneous marginal conditions into a symmetric system on an undirected graph. A potential and a spectral condition can then give uniqueness.

Several network layers

An agent chooses effort across multiple kinds of ties. A transformed system can have a potential even with cross-layer costs. Extra curvature conditions are needed for uniqueness.

The same method can reveal different mathematical structures across models.

CES peer effects on an unnormalized network

$$u_i(x) = a_i x_i - \frac{1}{2}x_i^2 + \delta x_i \Big(\sum\nolimits_j g_{ij} x_j^{\beta}\Big)^{1/\beta}.$$

Best response

$$T_i(x) = a_i + \delta S_i(x), \qquad S_i(x) = \Big(\sum\nolimits_j g_{ij} x_j^{\beta}\Big)^{1/\beta}.$$

- $a_i > 0$, $\delta > 0$, $\beta > 0$, and $x_i \geq 0$.
- G is an undirected adjacency matrix, without self-loops.
- We use a connected, nonempty network for the Perron-limit results.

What the CES parameter changes

For positive neighbor actions,

$$\frac{\partial S_i}{\partial x_j} = g_{ij} x_j^{\beta-1} S_i(x)^{1-\beta}.$$

For two linked neighbors j and k,

$$\frac{\partial S_i / \partial x_j}{\partial S_i / \partial x_k} = \left(\frac{x_j}{x_k}\right)^{\beta-1}.$$

Relative peer influence

$\beta > 1$ favors high-action peers at the margin. $0 < \beta < 1$ places relatively more marginal weight on low-action peers.

Changing β also changes the scale of this unnormalized aggregate.

CES complements: uniqueness without a closed form

Every equilibrium has $x_i > 0$. Write $z_i = \log x_i$ and compare the positive quantities in the best-response equation:

$$H_i(z) = z_i - \log(a_i + \delta S_i(e^z)) .$$

What the transformation reveals

$$DH(z) = I - W(z), \qquad W(z) \geq 0, \qquad \sum_j W_{ij}(z) = \frac{\delta S_i}{a_i + \delta S_i} < 1.$$

The positive baseline a_i leaves some own-action weight in every row. Hence there is at most one finite positive equilibrium.

Whether a finite equilibrium exists depends on the network threshold on the next slide.

The spectral existence threshold

Theorem 4

For $\bar{\delta} = \lambda_{\max}(G)^{-1/\beta}$:

- $0 < \delta < \bar{\delta}$: a unique positive equilibrium exists.
- $\delta \uparrow \bar{\delta}$: all actions diverge, with convergent ratios.
- $\delta \geq \bar{\delta}$: no finite positive equilibrium exists.

On a d-regular network with $a_i = a$,

$$x_i^* = \frac{a}{1 - \delta d^{1/\beta}}, \qquad \bar{\delta} = d^{-1/\beta}.$$

Comparative statics below the threshold

How shocks change behavior

Within the region where a finite equilibrium exists,

$$a' \geq a \quad \Longrightarrow \quad x^*(a', \delta, \beta) \geq x^*(a, \delta, \beta),$$

$$\delta' \geq \delta \quad \Longrightarrow \quad x^*(a, \delta', \beta) \geq x^*(a, \delta, \beta).$$

- A higher individual baseline or stronger peer interaction raises equilibrium effort across the network.
- Equilibrium payoff is $u_i(x^*) = \frac{1}{2}(x_i^*)^2$: changes in effort also change welfare.

Bonacich centrality as the linear special case

When $\beta = 1$,

$$x = a + \delta G x.$$

The network multiplier

$$(I - \delta G)x = a, \qquad x^* = (I - \delta G)^{-1} a.$$

- $a = \mathbf{1}$: a standard Katz–Bonacich centrality vector.
- Heterogeneous a: baseline-weighted centrality.
- In the notation $(I - \delta G)x = b$, the vector b is this same baseline.

Eigenvector centrality at the critical limit

For the positive unit Perron vector e,

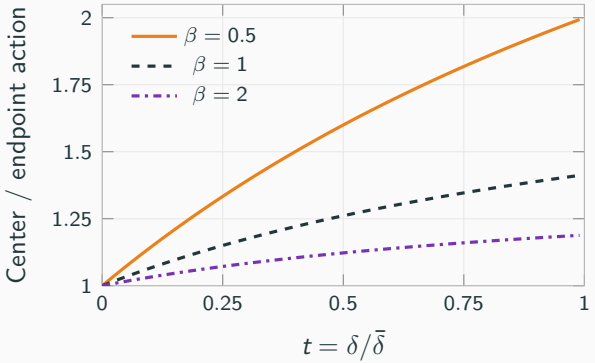
$$Ge = \lambda_{\max}(G)e, \qquad e \gg 0.$$

Theorem 4(ii)

$$\lim_{\delta \uparrow \bar{\delta}} \frac{x_i^*}{x_j^*} = \left(\frac{e_i}{e_j}\right)^{1/\beta}.$$

- $\beta = 1$: $x^*/\|x^*\|_2 \to e$.
- General $\beta > 0$: the vector $(x^*)^\beta$ approaches the Perron direction.
- Baseline heterogeneity affects levels, but disappears from these limiting ratios.

Relative actions along the path



$$\frac{C}{L} = \frac{1 + tq}{1 + t/q}, \qquad q = 2^{1/(2\beta)}.$$

As $t \uparrow 1$

The ratio converges to q.

The action levels diverge.

At the same fraction of the threshold, smaller β produces a larger center advantage here.

The key-player intervention

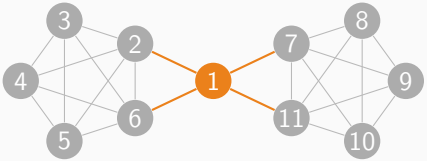
Let G^{-i} denote the network after removing player i.

Total equilibrium activity lost

$$\Delta_i = \sum_j x_j^*(G) - \sum_{j \neq i} x_j^*(G^{-i}), \qquad i^* \in \arg\max_i \Delta_i.$$

- Removing a player eliminates their action and changes everyone else's incentives.
- Highest equilibrium action need not identify the largest total effect.
- For general β, recompute equilibrium on each remaining network.

A bridge can matter more than the most active player



Linear benchmark, $\beta = 1$

δ	Key-player type
0.1	Node 2
0.2	Node 1

Nodes 2, 6, 7, and 11 have the highest degree and the highest action.

Nonlinear peers change the intervention ranking

At $\delta = 0.1$, the paper finds a switch from node 1 to the node-2 type near $\beta = 0.722$ within its reported parameter range.

Network public goods with nonlinear costs

On an undirected network, player i chooses $x_i \geq 0$ and receives

$$u_i(x) = b\left(x_i + \delta \sum_j g_{ij} x_j\right) - c(x_i), \qquad 0 < \delta \leq 1.$$

Assumption 2: benefits, costs, and a marginal crossing

$$b' > 0, \qquad b'' < 0, \qquad c'(x) > 0 \ (x > 0), \qquad c'' \geq 0,$$

$$b'(0) > c'(0), \qquad \lim_{x \to \infty} b'(x) < \lim_{x \to \infty} c'(x).$$

The positive autarky effort k^* uniquely solves $b'(k^*) = c'(k^*)$.

The benchmark of Bramoullé and Kranton (2007) has $c(x) = x$ and $\delta = 1$.

Best responses, corners, and existence

Let $t_i = \delta \sum_j g_{ij} x_j$ be the public good supplied by neighbors.

If $x_i > 0$

The agent balances marginal benefit and cost:

$$b'(x_i + t_i) = c'(x_i).$$

If $x_i = 0$

Neighbors may already provide enough that contributing the first unit is not worthwhile.

More provision by neighbors weakly reduces one's best response. Every best response is at most the autarky effort k^*, so a finite equilibrium exists.

Substitutes: curvature can restore uniqueness

The original marginal conditions are asymmetric because each agent receives different total provision. A sign-equivalent inverse transformation gives

$$\widetilde{F}_i(x) = x_i + \delta \sum_j g_{ij} x_j - h(x_i), \qquad h = (b')^{-1} \circ c'.$$

Economic result

On an undirected graph this system has a potential. The sufficient condition

$$1 + \delta \lambda_{\min}(G) > 0$$

ensures a unique equilibrium. Convex private costs add curvature and can support uniqueness beyond this simple bound.

The sign equivalence also retains equilibria with zero effort.

Zero marginal cost at zero implies full participation

Now suppose c is strictly convex, $c'(0) = 0$, and $\delta = 1$, as in Proposition 7.

Every equilibrium is interior

At any finite profile with $x_i = 0$,

$$\left.\frac{\partial u_i}{\partial x_i}\right|_{x_i=0} = b'\Big(\sum\nolimits_j g_{ij}x_j\Big) - c'(0) > 0.$$

Hence $x_i^* > 0$ for every player at every Nash equilibrium.

All equilibrium conditions therefore hold with equality:

$$b'\Big(x_i^* + \sum\nolimits_j g_{ij}x_j^*\Big) = c'(x_i^*).$$

Full participation does not require equilibrium uniqueness.

Complete networks: convex costs remove the continuum

Take the complete network and $\delta = 1$. Every player receives total provision $S = \sum_j x_j$.

Strictly convex costs, $c'(0) = 0$

$$c'(x_i) = b'(S) \quad \text{for all } i.$$

Strictly increasing c' forces equal effort.

$$x_i = \bar{x}, \qquad b'(n\bar{x}) = c'(\bar{x}).$$

The equilibrium is unique.

Linear costs, $c(x) = x$

With $b'(k^*) = 1$, every profile satisfying

$$x_i \geq 0, \qquad \sum_i x_i = k^*$$

is an equilibrium.

The continuum includes both equal sharing and a single specialist.

Here $\lambda_{\min}(G) = -1$. Uniqueness with strictly convex costs holds even though $1 + \delta\lambda_{\min}(G) = 0$.

Network structure determines effort rankings

Maintain homogeneous b, c, strictly convex costs, $c'(0) = 0$, and $\delta = 1$. Let N_i and $\overline{N}_i = N_i \cup \{i\}$ denote open and closed neighborhoods.

Neighborhood inclusion

At every equilibrium,

$$N_i \subsetneq N_j \text{ or } \overline{N}_i \subsetneq \overline{N}_j \implies x_i^* > x_j^*.$$

- **Star, $n \geq 3$.** Leaves choose equal effort. The hub contributes less.
- **Nested split graph.** If $d_i < d_j$, then $x_i^* > x_j^*$.

Degree alone does not imply this ranking on arbitrary networks.

What SET establishes in these applications

Question	Required argument
Equilibrium equivalence	Valid VI representation, full-domain sign equivalence, and appropriate domain geometry.
Existence	A compactness, coercivity, strong-P, or bounded-iteration argument.
Uniqueness	A suitable P property, strict monotonicity, or strict convexity.

Main conclusions

A general method

On rectangular domains, the signs of the equilibrium operator determine the VI solution set. Suitable transformations can reveal structure absent from the original Jacobian.

- CES complements combine a SET uniqueness argument with a Perron existence threshold.
- Bonacich centrality is a resolvent response. Eigenvector centrality describes its critical direction.
- Under strategic substitutes, cost curvature and the smallest network eigenvalue govern uniqueness.

Which transformation exposes the useful structure in a given model?

Backup: deriving the production unit cost

The normalized quantity technology corresponding to the main slides is

$$m_i = \left(\sum_j \omega_{ij}^{1/\sigma_i} q_{ij}^{1-1/\sigma_i} \right)^{\sigma_i/(\sigma_i-1)}, \qquad y_i = z_i \left(\frac{l_i}{\alpha_i} \right)^{\alpha_i} \left(\frac{m_i}{1-\alpha_i} \right)^{1-\alpha_i}.$$

Here q_{ij} is the quantity of input j, l_i is labor, and m_i is the bundle. The minimum cost of m_i is $Q_i(p)m_i$.

For a target output, let $E_i = wl_i + Q_i m_i$. Cost minimization gives

$$wl_i = \alpha_i E_i, \qquad Q_i m_i = (1-\alpha_i) E_i.$$

Substituting the optimal input quantities into the technology yields

$$y_i = z_i \left(\frac{E_i}{w} \right)^{\alpha_i} \left(\frac{E_i}{Q_i} \right)^{1-\alpha_i} = \frac{z_i E_i}{w^{\alpha_i} Q_i^{1-\alpha_i}}.$$

Therefore $c_i = E_i/y_i = w^{\alpha_i} Q_i^{1-\alpha_i}/z_i$.

Backup: the production-network matrix test

For $\sigma_i > 1$, the log-price map and its derivative are

$$T_i(x) = -\varepsilon_i + \frac{1-\alpha_i}{1-\sigma_i} \log\left(\sum_j \omega_{ij} e^{(1-\sigma_i)x_j}\right),$$

$$W_{ij}(x) = \frac{\partial T_i}{\partial x_j} = (1-\alpha_i)\frac{\omega_{ij} e^{(1-\sigma_i)x_j}}{\sum_k \omega_{ik} e^{(1-\sigma_i)x_k}}.$$

The fraction is supplier j's share within intermediate expenditure. Multiplying by $1-\alpha_i$ gives its share in total cost.

$$W \geq 0, \qquad \sum_j W_{ij} = 1-\alpha_i, \qquad D(x - T(x)) = I - W(x).$$

The diagonal dominance margin of $I - W$ in row i equals $\alpha_i > 0$. Every principal submatrix has positive determinant, so $I - W$ is a P-matrix. The uniform margin $\min_i \alpha_i > 0$ also yields the contraction used in the main slides.

Backup: a formal coordinate proof of Theorem 1

Let $K = \prod_i K_i$ and suppose $x^* \in \mathrm{Sol}(K, F)$. For every i and $t \in K_i$,

$$(t - x_i^*) F_i(x^*) \geq 0.$$

Sign equivalence implies the same inequality for $\widetilde{F}_i(x^*)$:

$$\begin{cases} F_i(x^*) = 0 & \Longrightarrow \ \widetilde{F}_i(x^*) = 0, \\ F_i(x^*) > 0 & \Longrightarrow \ t - x_i^* \geq 0 \text{ and } \widetilde{F}_i(x^*) > 0, \\ F_i(x^*) < 0 & \Longrightarrow \ t - x_i^* \leq 0 \text{ and } \widetilde{F}_i(x^*) < 0. \end{cases}$$

Consequently, for every $x \in K$,

$$\langle x - x^*, \widetilde{F}(x^*) \rangle = \sum_i (x_i - x_i^*) \widetilde{F}_i(x^*) \geq 0.$$

Thus $\mathrm{Sol}(K, F) \subseteq \mathrm{Sol}(K, \widetilde{F})$. Interchanging the two operators gives the reverse inclusion.

Backup: matrix tests in the payoff-scaling example

For $J = \begin{pmatrix} 1 & -\delta \\ -\kappa\delta & \kappa \end{pmatrix}$ with $\kappa > 0$:

P-matrix test

The principal minors are

$$1, \quad \kappa, \quad \kappa(1-\delta^2).$$

All are positive if $0 < \delta < 1$.

Strict monotonicity test

For $S = (J + J^\top)/2$,

$$\det S = \kappa - \frac{(1+\kappa)^2\delta^2}{4}.$$

$S \succ 0$ iff $\delta < 2\sqrt{\kappa}/(1+\kappa)$.

Let $\Delta = (1-\kappa)^2 + (1+\kappa)^2\delta^2$. The smallest eigenvalue is

$$\lambda_{\min}(S) = \frac{1+\kappa-\sqrt{\Delta}}{2}.$$

Rescaling by $\mathrm{diag}(1, 1/\kappa)$ restores the symmetric derivative with eigenvalues $1-\delta$ and $1+\delta$.

Backup: the nonlinear Perron limit

Let $y(\delta) = x(\delta)^\beta$ and normalize $q(\delta) = y(\delta)/\|y(\delta)\|_1$.

- If actions remained bounded as $\delta \uparrow \bar{\delta}$, continuity would give an equilibrium at $\bar{\delta}$.
- Connectedness propagates divergence: $x_i \geq \delta x_j$ whenever $g_{ij} = 1$.

Therefore $a_i/x_i \to 0$, and along any convergent subsequence,

$$\frac{(x-a)^\beta}{\|x^\beta\|_1} = \delta^\beta G q(\delta) \quad \Longrightarrow \quad q = \bar{\delta}^\beta G q.$$

Uniqueness of the normalized limit

The only nonnegative unit-ℓ_1 Perron vector is $e/\|e\|_1$. Every subsequential limit is the same, so the full sequence converges.

Backup: algebraic solutions and positive equilibria

For two linked agents with baseline $a = \mathbf{1}$,

$$G = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \lambda_1 = 1, \quad \lambda_2 = -1.$$

Whenever $\delta \neq \pm 1$, the linear system has the solution

$$x = (I - \delta G)^{-1}\mathbf{1} = \frac{1}{1-\delta}\mathbf{1}.$$

Parameter	Algebraic system	Economic equilibrium
$0 < \delta < 1$	Unique solution	Positive and finite
$\delta = 1$	Singular, inconsistent	None
$\delta > 1$	Unique negative solution	None on $\mathbb{R}^2_+$

Backup: complete-network uniqueness without the strict spectral bound

Assume homogeneous b, c, strictly convex c, $c'(0) = 0$, and $\delta = 1$. On the complete network, write $S = \sum_j x_j$.

1. Full participation makes every first-order condition an equality:
$$c'(x_i) = b'(S) = c'(x_j) \qquad \text{for every } i, j.$$
2. Strictly increasing c' implies $x_i = x_j = \bar{x}$.
3. The function $f(z) = b'(nz) - c'(z)$ is strictly decreasing, with $f(0) > 0$ and an eventually negative value. Thus it has exactly one positive zero.

Why the spectral test is only sufficient

For $n \geq 2$, the complete graph has $\lambda_{\min}(G) = -1$. At $\delta = 1$ the strict spectral inequality fails, but the argument above proves uniqueness.

Backup: why the hub contributes less in a star

Let $n \geq 3$ and $\delta = 1$, with homogeneous functions, strictly convex costs, and $c'(0) = 0$. Write x_H for the hub's effort.

1. Every leaf faces the same peer effort x_H and has a unique best response. Thus all leaves choose one common effort $x_L > 0$.
2. The hub and a leaf satisfy
$$c'(x_H) = b'(x_H + (n-1)x_L), \qquad c'(x_L) = b'(x_H + x_L).$$
3. Since $(n-2)x_L > 0$ and b' is strictly decreasing,
$$c'(x_H) < c'(x_L).$$
Strictly increasing c' then gives $\boxed{x_H < x_L}$.

The proof applies to every equilibrium, without a uniqueness assumption.

References

- Zenou, Y. and Zhou, J. (2026). *SET*. Working paper, February 17, 2026.
- Ballester, C., Calvó-Armengol, A. and Zenou, Y. (2006). Who's Who in Networks. Wanted: The Key Player. *Econometrica*, 74(5), 1403–1417.
- Bramoullé, Y. and Kranton, R. (2007). Public goods in networks. *Journal of Economic Theory*, 135(1), 478–494.

The seminar follows the supplied SET draft. The two-player walkthroughs, two-industry production example, path-network curves, and two-node calculation are seminar examples.